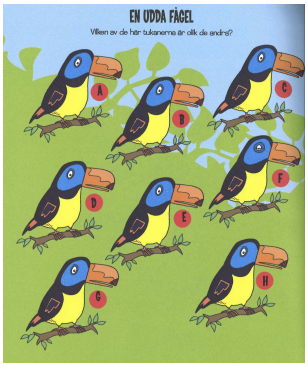
LUNDS TEKNISKA HÖGSKOLA
MATEMATIK

TENTAMENSSKRIVNING
Image Analysis
2026 HOME EXAM

Home Exam
You may use any books and computer programs (e.g. Python, Matlab and Maple), but it is not permitted to get help from other persons or dedicated AI-tools such as ChatGPT as detailed on the Canvas page for the course. It is ok to use existing implementations of known methods, but the methods used should be explained in detail. The exam should be uploaded via the Canvas homepage, at the latest
72 hours after pick up.

1. In the file `birds_exam.jpg` an image is given (also shown here to the right). One of the birds is different than the others. Construct a method that automatically finds the different bird.



2. a) A gray-level input image, $0 \leq I(x, y) \leq 1$ (using the continous model), is transformed using ordinary histogram equalization. After the transformation, the image is thresholded at gray-level 0.5. Assume now that the input image has a gray-level intensity distribution given by

$$p(r) = 3r^2, 0 \leq r \leq 1. \tag{1}$$

At what gray-level r should the input image be thresholded at, in order to get the same result as thresholding the output image at gray-level 0.5?

We will now study a variant of standard histogram equalization. In ordinary histogram equalization one can show that if we want a flat output grayscale distribution we should use the grayscale transformation defined by

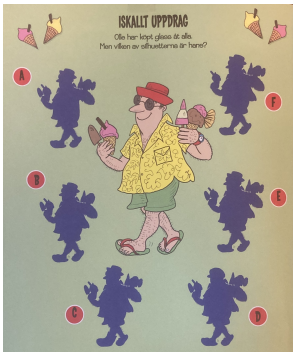
$$T(r) = \int_0^r p(t)dt, \tag{2}$$

where p is the input grayscale distribution of the image.
b) Now assume that we want the output distribution instead to be

$$p(s) = 2s, \tag{3}$$

how should we then define the grayscale transformation?
c) For the input distribution given by $p(r) = 6r(1-r)$, how should $T(r)$ be chosen so that the output distribution becomes $p(s) = 2s$?

3. In the file `icecream_exam.jpg` an image is given (also shown here to the right). A number of possible shadows are given where only one corresponds to the figure in the middle. Construct an algorithm that finds which one. Make your method as automatic as possible. In a second image (`icecream_exam_2.jpg`) a more difficult example is given. Try your method on this as well.



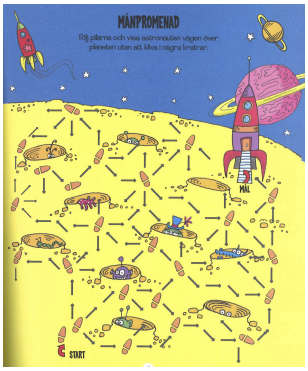


4. Assume that we are given the task of training a classification method using a CNN. The input is 32×32 grayscale images, and we want to classify them into 5 classes. We have a CNN-model with the following layers: Convolution layer with 3×3 kernel and 16 output channels (stride 1 and padding 1), A RelU layer, a 4×4 Max pooling layer with stride 4, a Convolution layer with $3 \times 3 \times 16$ kernel and 32 output channels (with stride 1 and padding 1), A RelU layer, a 2×2 Max pooling layer with stride 2, a fully connected layer from $4 \times 4 \times 32$ input nodes to 5 output nodes, and finally a Softmax layer. a) Calculate how many parameters we need to find and store for this model. b) The receptive field of a feature in a specific layer is defined as the size of the region in the input image that affects the feature. What is the receptive field of the features in the final max pooling layer? Here you don't have to consider edge effects at the image border (note also that all the kernels are symmetric in size in the $x-$ and $y-$direction, so you only need to reason in one dimension).

5. In the files `cars_sm_1.png` and `cars_sm_2.png` two images of a scene are given (shown to the right). The second image is corrupted and only contains the last two color channels (i.e. the values in the red channel are all zero). Find the correct colors by estimating the red channel in some automatic way.




6. In the file `moon_exam.jpg` an image is given (also shown here to the right). Find the way from the start to the rocket (MÅL). You may only move along the arrow directions, and you should avoid the craters. Make your method as automatic as possible.



Good Luck!